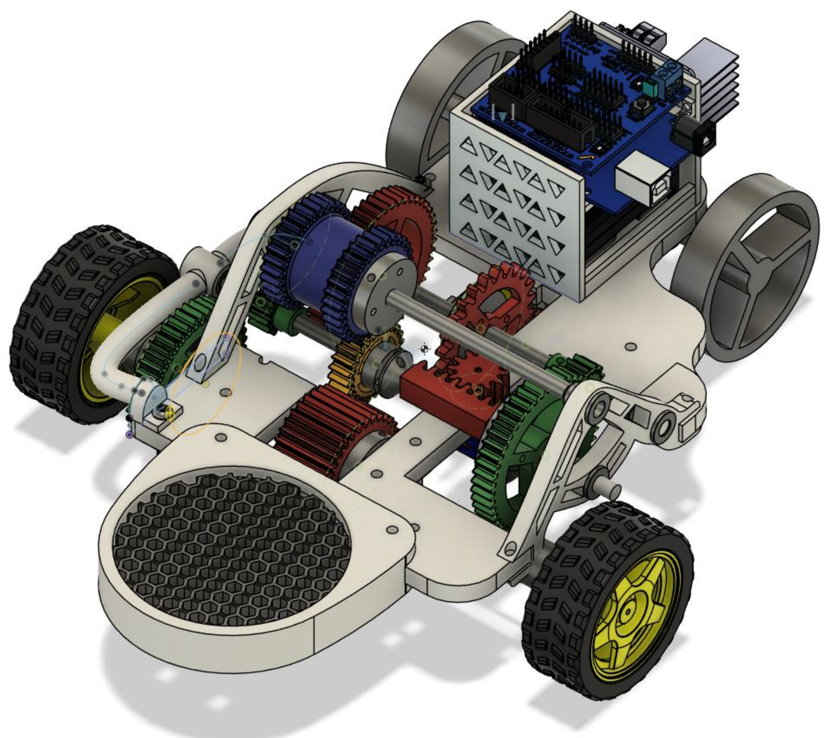
ME 371 Final Report



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Team 14

December 9, 2025

Professor Wandke

I. Overview

The remote-control mobile stage prototype was designed to strategically maximize score in the heptathlon by targeting reaching speed, strength, and agility goals. To reach the necessary range of speeds/torques, a compound gear train was utilized for the high torque speed while the reverse and high-speed ratios relied on a simple gear train with one gear on each intermediate shaft. A floating idler gear is used to implement a reverse function. Shifting is done with a sliding intermediary shaft. A key design consideration in our shifter was allowing the rack to freely spin on the shaft but using collars that would allow for the translation of linear motion with the rack and pinion mechanism. In figure 1 below, the shaft collars used to secure the rack linearly are shown in white.

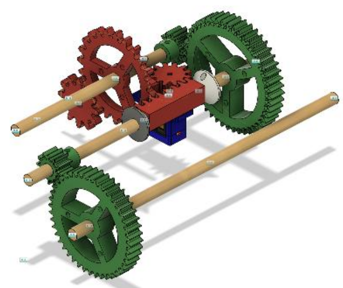


Figure 1 Cad model of high torque shifting mechanism

The drivetrain was designed with relative gear meshing distances of 15 mm in each direction with the middle servo position meshing with the idler gear, utilizing the reverse function. Then, a calculation was done to calculate the length of the rack and the size of the pinion:

$$\boldsymbol{Pitch\ Diameter_{servo\ gear} = \frac{15 * 4}{\pi} = 19.1\ mm} \quad (1)$$

The rack and pinion size were validated in the Fusion360 [1] model for size and packaging constraints. With one servo, the shifting mechanism shifts through three speeds designed at the servo's extrema and neutral (middle) positions. The default code is used to shift the car with a specific pattern that was practiced for the agility test. The gear train speeds are summarized in table 1, where the detailed calculation process used to obtain these gear ratios is addressed in Appendix C.

Table 1. Gear train ratios for all speeds

Forward	1:1.5
Reverse	1:0.75
High Torque	29.33:1

The theme of the robot is inspired by the Sydney Opera House. The curves, irregular shapes, and walls of the architecture were translated into the walls, colors, and asymmetry of the design. A stage is placed at the back of the chassis for the durability test to absorb the impact.

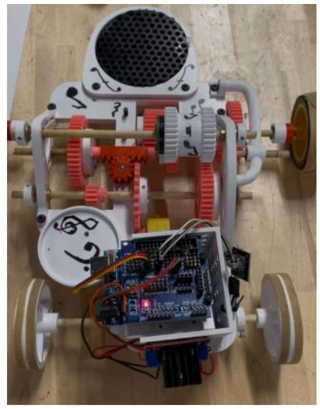


Figure 2. Top-down view of CAD model with Stage design

The high torque is achieved with a compound gear ratio. Inspecting figure 2, starting from the bottom, the first axle has 3 gears: a gear in constant mesh with the idler gear, the gear that is in constant mesh with the motor input, and a small high torque gear. Looking at the next set of axles, one can see a corresponding high torque gear and a small gear on the other end. A better view is seen in figure 3. On the elevated axle is the blue idler gear. Figure 3 demonstrates the reverse function of the transmission. As the yellow gear shifts towards the bottom of the page, it releases from the idler gear, and the small green high torque gear applies rotation to the large green output gear towards the bottom of the page. The blue idler gear is designed with an irregular shape, so the input shaft has constant mesh. The driven axle also has two gears: one for high torque and the other, long one for forward as the shifting axle slides towards the top of the page and disengages with the idler gear.

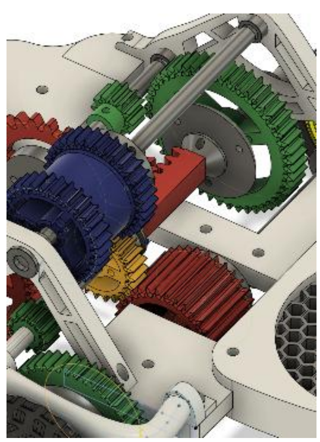
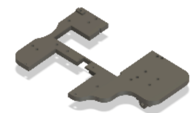
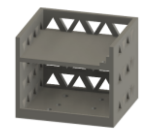
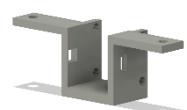
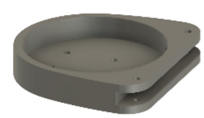
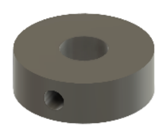
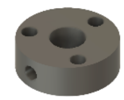
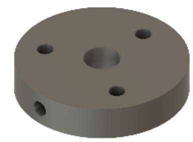
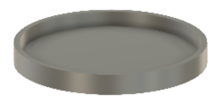
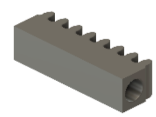


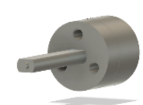
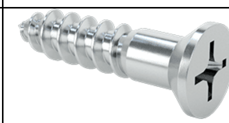
Figure 3. Isometric zoomed in view of Model

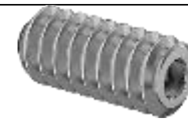
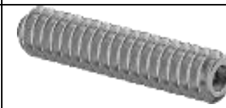
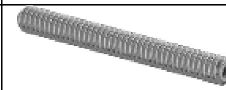
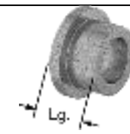
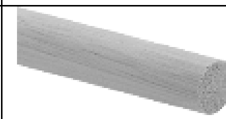
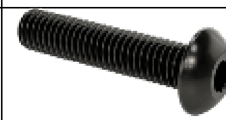
II. Bill of Materials

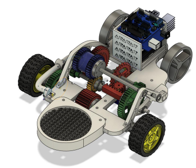
Table 2. Bill of Materials

Part	Total Mass/count	Source	Price ($)	Image
PLA	277g	JIS	8.31	      

				      
TPU	25g	JIS	1.25	
Flathead #4, 0.75” (2)	21	JIS	1.176	

Set Screw 4-40 1/4”	7	JIS	0.581	
Set Screw 4-40 ½”	4	JIS	0.5	
Set Screw 4-40 1”	3	JIS	0.501	
Flange Bearing ¼” SD ½”	5	JIS	10	
Sleeve Bearing ¼” SD ½”	1	JIS	2	
¼” Wooden Dowel	4	JIS	2	
M4 Bolt	10	JIS	1.67	
Arduino Board	1	Kit	0	
Motor Control Board (Red Board)	1	Kit	0	
IR Receiver	1	Kit	0	
IR Remote	1	Kit	0	
DC Motor	1	Kit	0	
Battery Pack	1	Kit	0	
SG 90 Servo Motor	1	Kit	0	
Caster Wheel	1	Kit	0	
Tire Wheel	2	Kit	0	

		Grand Total	27.98	

III. Machine Component Analysis

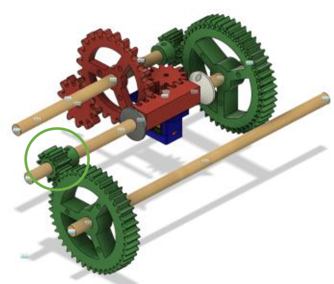


Figure 4 Machine Component Analysis Part Selection

For the Machine Component Analysis, the smallest gear with the highest force is inspected for failure. The component is key to giving the robot high torque functionality and is seen in figure 4. The component is made of 25% grid infill PLA, and the specifications are taken from the sheet on Canvas "References for PLA Analysis" [2]. The component's key features, geometry, and material properties are listed in table 3.

	Value
Young's Modulus	2.76 GPA
Yield Strength	28.1 MPa
Ultimate Strength	29.7 MPa
Gear Shape	Spur
Number of teeth	12
Module (m)	1.25 mm
Radius	15 mm
Pressure angle	20°

Table 3. Key properties of part selected for analysis

Then, the worst-case scenario is assumed for the operating condition of the motor: 200RPM minimum speed and 0.8 kg.cm stall torque. Equation for stress in a gear tooth fillet due to bending,

$$\boldsymbol{\sigma = \frac{F_t P}{bJ} K_v K_o K_m,} \tag{2}$$

can be used to estimate the stress in the teeth. A free body diagram of the gear under load is given in figure 5. F_N is the normal force that acts perpendicular to the surface of the mating gear, but we are only interested in F_t since the teeth are the weakest component of the gear and most likely to fail.

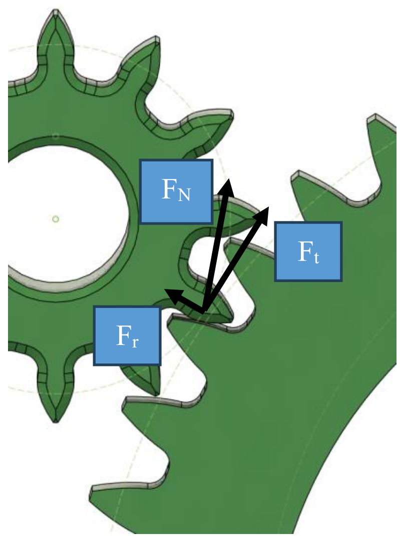


Figure 5 Free Body diagram of selected component

This gear experiences 8 times the torque of the input torque (motor torque) with 12 teeth and diameter 15 mm (see appendix C for full calculations). P is simply calculated to be 800 teeth per meter of pitch diameter. Then figure 6 is used to find the geometry factor J at 20° pressure angle full depth teeth.

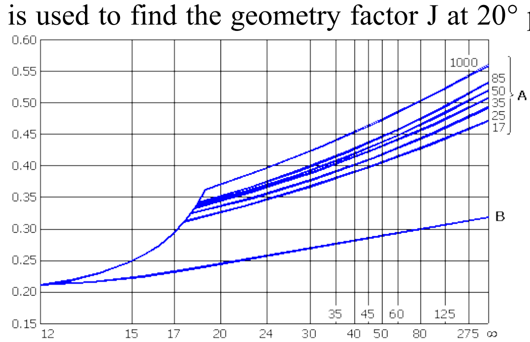


Figure 6 6 Geometry factor J for number of teeth vs number of teeth in mating gear with 20° pressure angle [8]

Since the gear has 12 teeth, the y-intercept is used for the geometry factor, J = 0.21. The velocity correction factor is calculated to be 1.013 (full calculations in appendix C). The overload factor is assumed to be 1.25 to account for the starting and stopping of the DC motor. A mounting correction factor is then assumed to be 1.5 for worst case scenario where the sliding mechanism is not aligned well with the mating gear. The tooth profile thickness, b = 10 mm. Plugging in tangential force = 83.68 N (see appendix C) to the equation,

$$\boldsymbol{\sigma = \frac{83.68 * 800}{0.01 * 0.21} * 1.013 * 1.25 * 1.5 = 60.78\ MPa} \tag{3}$$

This is twice the ultimate tensile stress of the PLA reference values. It can be said that when the motor is operating in this worst-case regime, the high torque gear will fail under repeated use. For a less conservative estimate of the stress, I assumed the motor to be operating at 75% of full capacity. This results in tangential force of 62.76 N. Plugging this into the equation,

$$\boldsymbol{\sigma = \frac{62.76 * 800}{0.01 * 0.21} * 1.013 * 1.25 * 1.5 = 45.41\ MPa} \tag{4}$$

This is still over the ultimate tensile stress of the PLA; however, when conducting the weight test, we noticed the motor stalling and the gears slipping without the gear teeth bending. This may be because the shaft collar is printed directly to the gear which may give the teeth additional structural support. There are also unaccounted losses in the transmission that are very likely reducing the torque at the small gear.

IV. FEA Stress Analysis

FEA analysis was conducted on the Left Wall Support Structure (Figure 7). Some of the key features of the wall are shown in Table 4. This wall support sits on the edges of the chassis and holds up the three main gear-shafts for our drivetrain. A key feature of this wall is the extended shaft holder (Hole 2, Figure 7). The purpose of this extension is to give enough clearance for the 12 toothed gear to slide 15 mm past the rest of the wall.

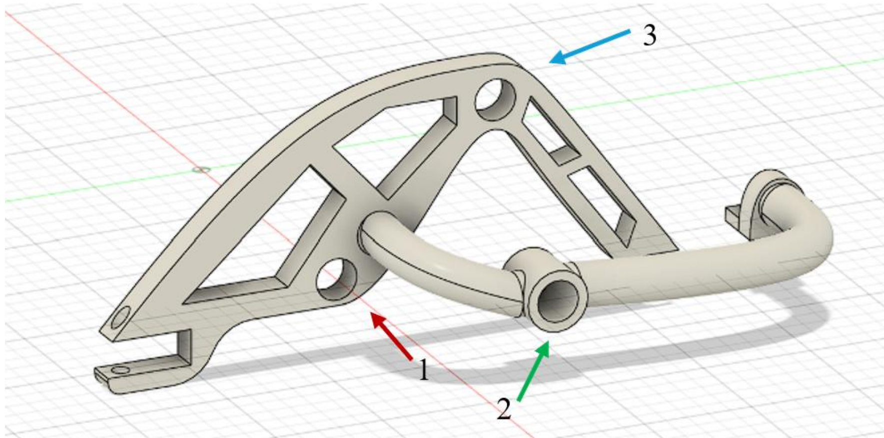


Figure 7. Left Wall Support for FEA Analysis, with Holes 1, 2, and 3

Table 4. Key Geometries and Properties of Left Wall Support

Geometries [mm]					**Tensile Properties for 25% Grid Infill PLA**		
Max Length	**Max Height**	**Max Width**	**Large Diameter**	**Small Diameter**	**Young's Modulus [GPa]**	**Yield Strength [MPa]**	**Ultimate Strength [MPa]**
162.4	61.31	42.64	9.525	4.000	2.76	28.1	29.7

For the stress analysis, the following loads were applied (Figures 8a, 8b, 9). The forces and moments acting downward on the holes represent the weights of the shafts and gears corresponding to each hole. For hole 1, a weight of 0.2 kg was estimated to be a distance of 90 mm away from the hole (Figure 8a). For hole 2, a weight of 0.35 kg was estimated to act 120 mm away from the hole (Figure 8b). Lastly, a weight of 0.1 kg acted 35 mm away from the hole (Figure 9). These distance values were found by finding the center of mass of the shafts while the sliding shaft is in high torque positioning, which would be the car's worst-loading case (with highest forces acting on the shafts). The bearing moments on each hole were estimated using the same equations and conditions applied in the machine component analysis (Appendix C.)

Figure 8a. FBD Analysis of Hole 1 -– 1.96 N Force, 21.0 N Bearing Moment, 0.177 Nm Bending Moment.

Figure 8b. FBD Analysis of Hole 2 – 3.43 N Force, 84.0 N Bearing Moment, 0.4116 Nm. Bending Moment

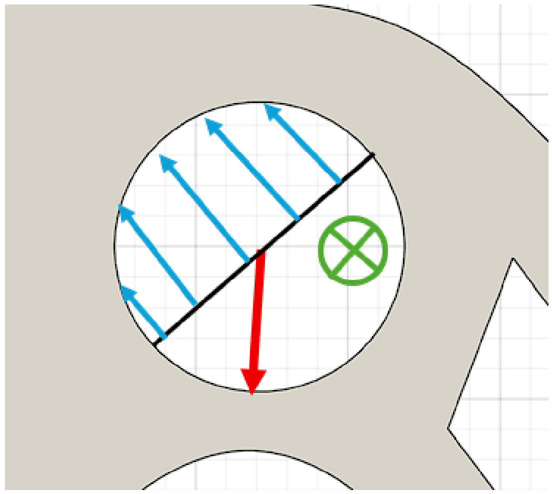


Figure 9 FBD Analysis of Hole 3 – 0.981 N Force, 20.0 N Bearing Moment, 0.0343 Nm Bending Moment

The smaller holes in the wall were designed to hold the M4 bolts, connecting the wall to the chassis. For our analysis, these 4 mm holes were constrained as fixed positions, and bearing loads were applied to the 3 shaft holes. With these boundary conditions, the following analysis was conducted.

The safety factor results (based on yield strength) indicated stress responses inside each hole that fell between a factor of 2 and 4 (Figure 10). However, the screw hole on the parabolic-shaped extrusion had a minimum safety factor of 0.854. The small thickness of the screw plate is the most likely cause of the lower factor of safety. For future consideration, this connector could have been a 2-sided connection (similar to the other screwed sections), so that more area could disperse the stresses at this point.

Next, the Von Mises graph shows that most of the support experienced a stress of 19.766 MPa or lower (Figure 11). A closeup of the parabolic section of the wall indicates a maximum stress region of 32.932 MPa, which is slightly above both the yield strength and tensile strength (Figure 12). When operating on the car, the team noticed this specific screw hole shearing off after multiple cycles of being screwed into and screwed out of the chassis. While this shearing did not noticeably impact the rigidity of the wall overall, nevertheless it validates the FEA model.

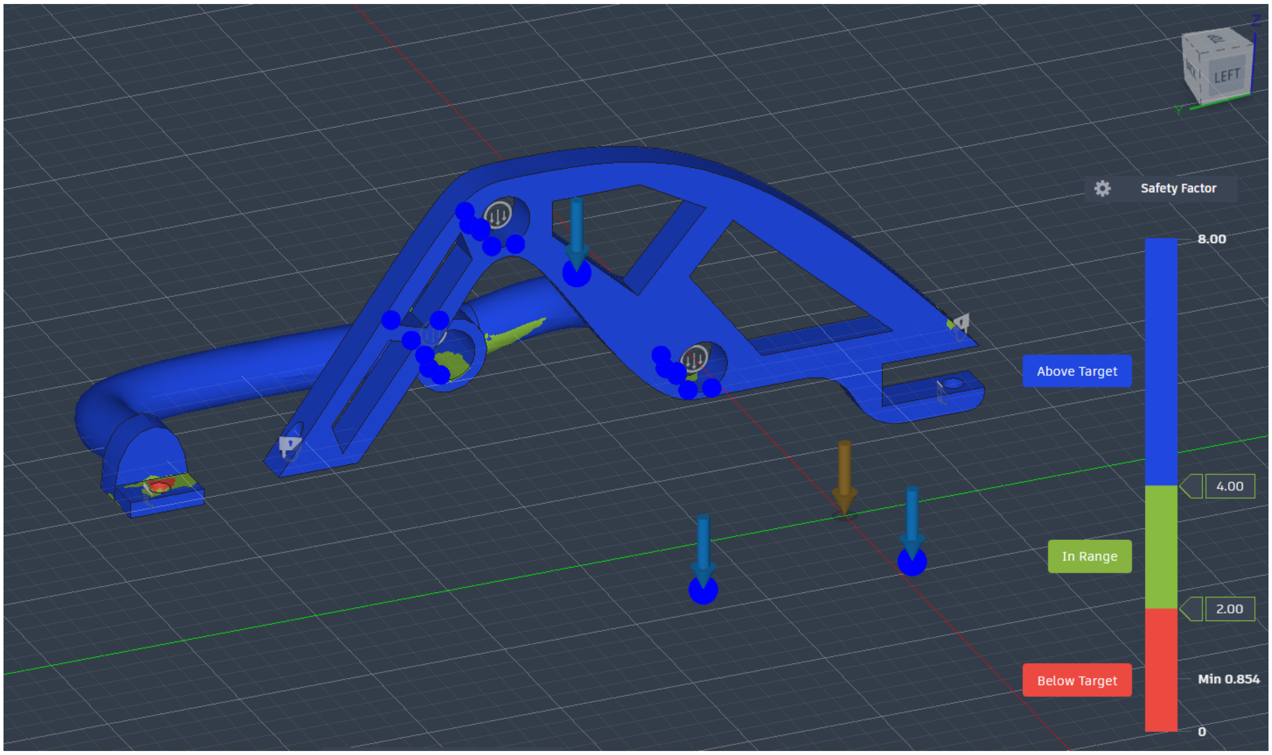


Figure 10. Gradient Map of Safety Factor

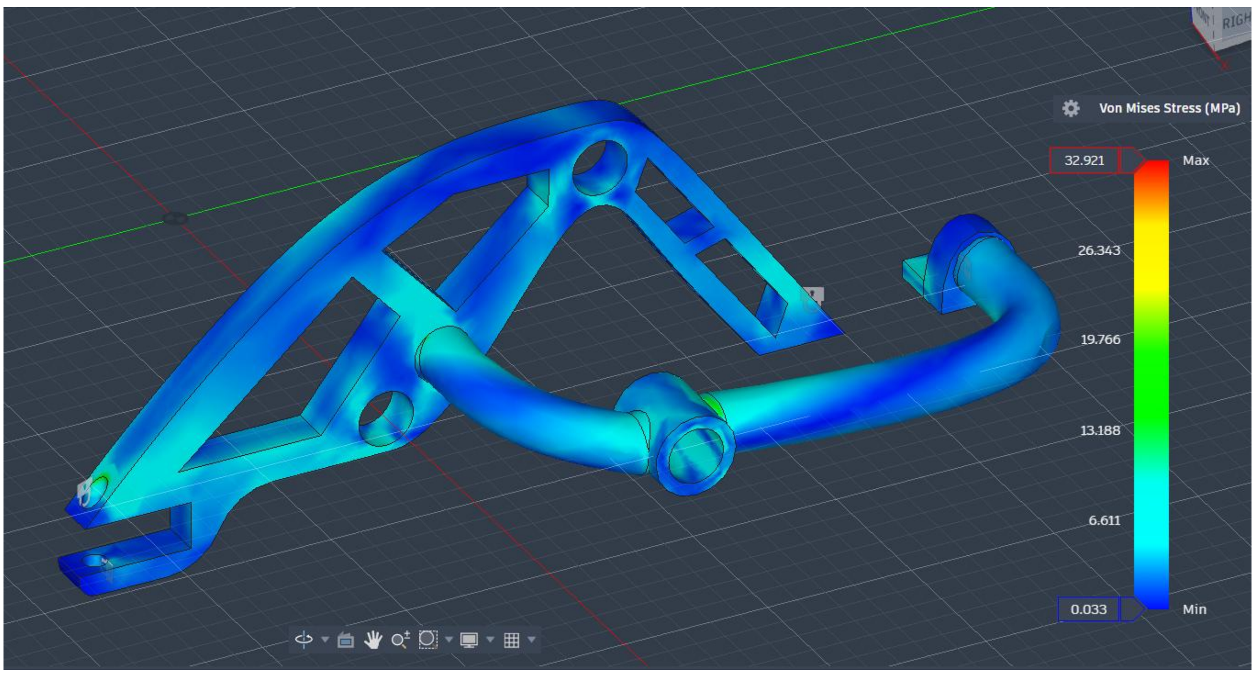


Figure 11. Gradient Map of Von Mises Stress

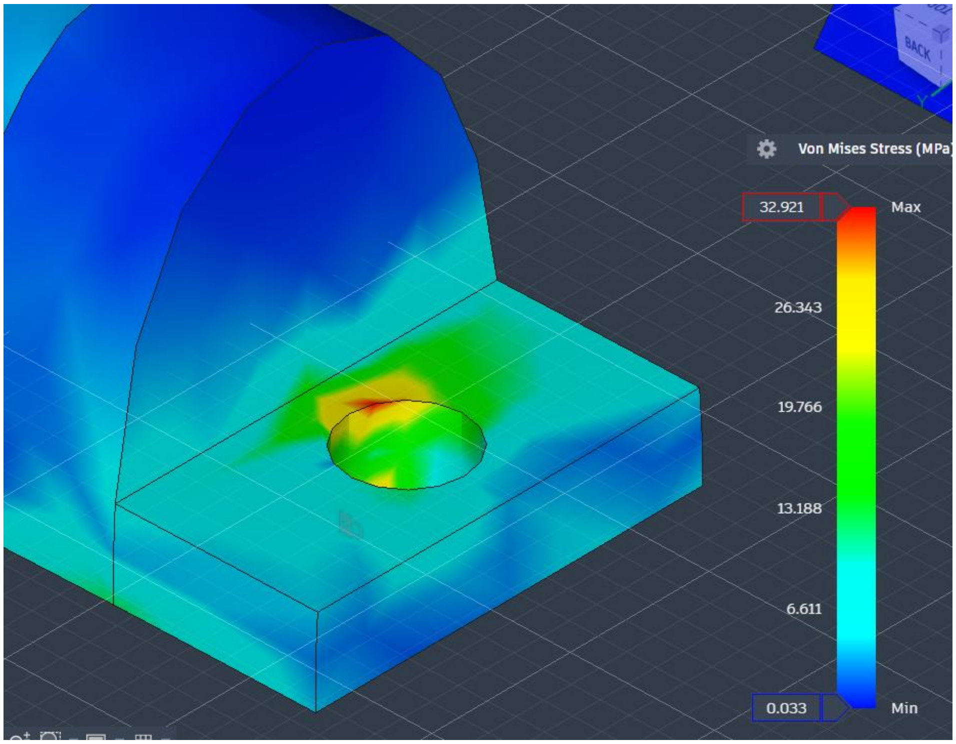


Figure 12. Zoomed In Gradient Map of Von Mises Stress

The displacement of the piece was mapped (Figure 13). The maximum displacement experienced was 1.155 mm, primarily around hole 2. Hole 3, the top of the support, and most of the tube extrusion holding hole 3, all tended to bend out of the screen.

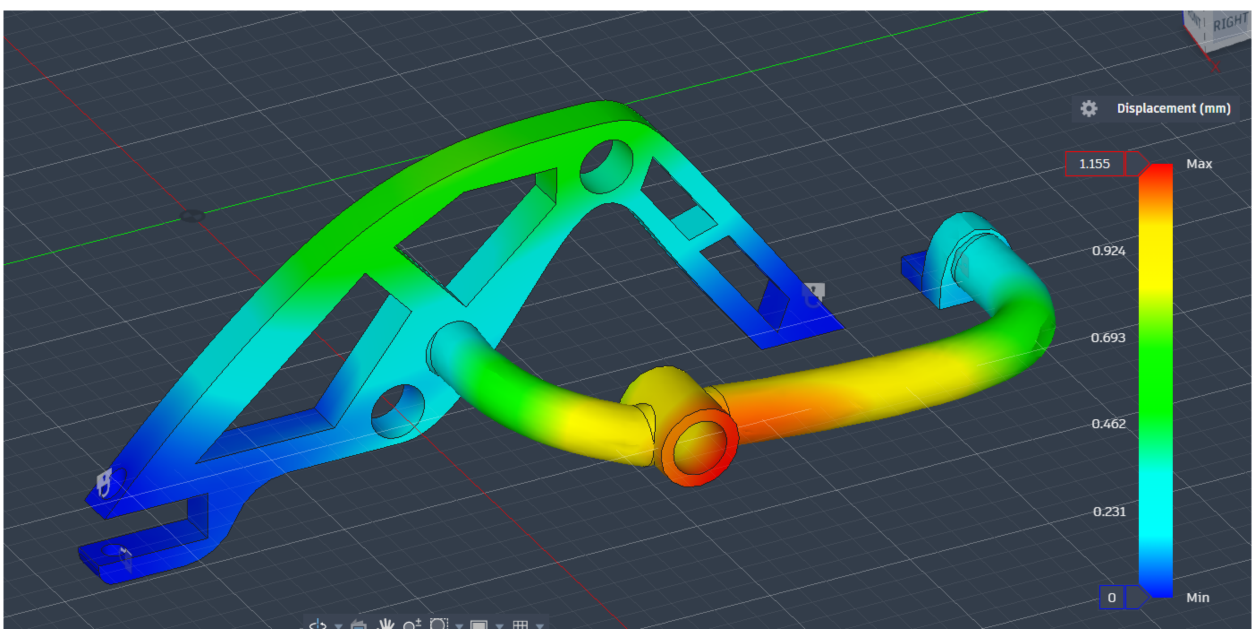


Figure 13. Gradient Map of Displacement

Mesh refinement was applied to the left wall as well (Figure 14). While most of the mosh covered the same shapes for the wall and tube-supports, it recommended removing material around parabolic-shaped chassis connection. This is most likely the case since that region did not experience much stress or displacement. For future consideration, team 14 would investigate removing this parabolic region, as well as repeating a FEA analysis on this wall support without cut-out sections to find out how the simulation would recommend removing material.

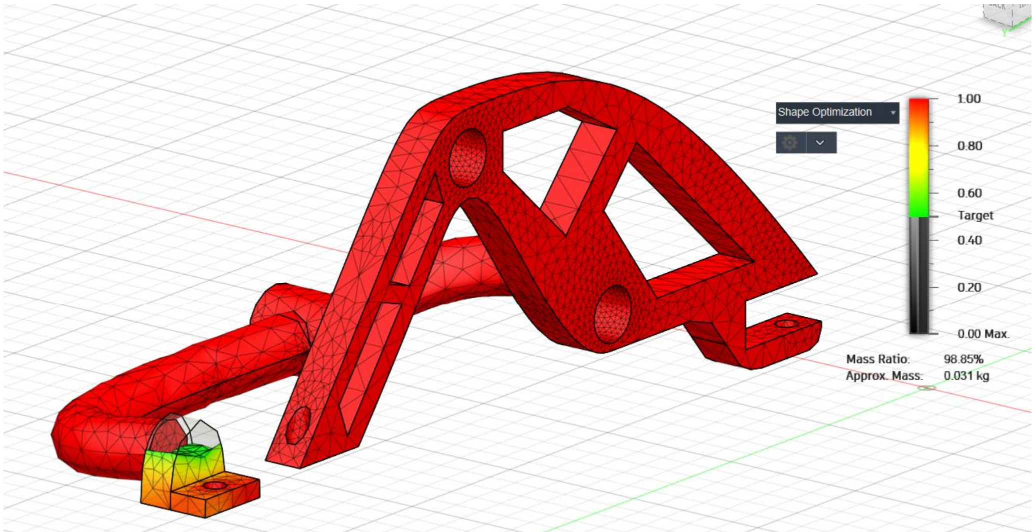
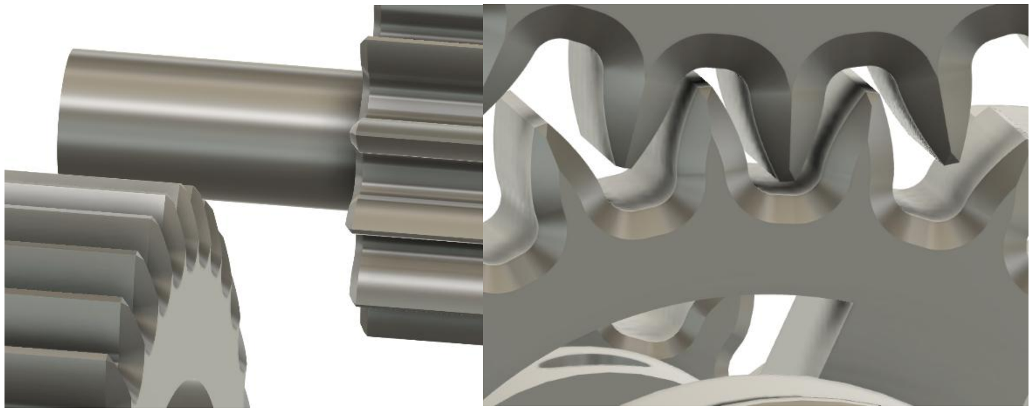


Figure 14.14 Refined Mesh of Left Wall

V. Reflection

The first major design decision is to implement a sliding-mesh shifting mechanism that incorporates a shaft sliding axially while being supported by two bearings. This decision is made to avoid controlling the motion of smaller components, for example, a clutch that the team considered hard to deal with. Instead, the motion of a larger rigid component, in this case, a shaft, is much easier to control and actuate with a servo motor. The tradeoff, however, is that all gears must be mounted to their exact position, with a tolerance of less than 2 mm of axial clearance between the gears. This became especially difficult when the team decided to use press-fit plus couplers (with set screws). Another design choice related to shifting is that the gears that are subject to disengagement and re-engagement from mesh have a chamfered surface and backlash in the gear teeth profile. This is to avoid chances of gear teeth clashing due to improper alignment, which is demonstrated in Figure 15. However, this could potentially lead to less efficient power transmission from the motor to the wheels, because the gears are now meshing with thinner teeth. Lastly, one more design choice the team made is to include a stage at the back of the car, supported by a caster wheel, and has a layer of TPU cushion to further absorb impact energy. The caster wheel is made of steel, which the team considered to be the strongest part.

When it comes to the biggest challenge throughout the project, the team encountered a lot of difficulties while handling the axial clearance between the gears. The gears must be mounted onto the



shaft precise to a millimeter to avoid double-meshing or half-meshing, since the distance of the shaft sliding is fixed. The team managed to solve this by careful labeling of gear positions, extensive trial and refinement, and adjusting the position of the pinion gear meshed onto the rack. It was an important lesson for the team that the servo's commanded angle must remain synchronized with the mechanical position of the sliding gear. Each servo state (–90°, 0°, +90°) corresponds to a specific gear engagement, so the pinion gear must be indexed very carefully on the rack so that it matches the gear's actual mechanical configuration.

Figure 15 Demonstration of meshing tolerances

The team iterated through the gear configurations multiple times to obtain the final gear layout by tackling and balancing between root fillet radius, modulus, and number of teeth. The root fillet radius corresponds to the size of the teeth, which directly impacts the power transmission rate. Larger teeth, in general, transmit power better. The number of teeth on each gear determines the gear ratio that the team has already calculated. Lastly, the modulus dictates the size of the gear, where in the original design, the gear sizes were so large that the axle-to-axle distance was more than 210 mm, but just having 9:1, 1:2.2, and 1:3 gear ratios. After careful design iteration, the team was able to shrink the axle-to-axle distance down to around 62 mm, reaching a more versatile and high-performance 1:0.75, 1:1.5, and 29.33:1 gear ratios. The team had to balance between the three features described that gears cannot excel in all three; compromises had to be made for the integral performance.

The design was tested primarily by running physical tests of the heptathlon objectives. The benchmark rubrics were very helpful in the development process, where in early stages the car reached intermediate performance milestones that focused on the fundamental functionalities. The team benefited from the qualifying round, where key performance statistics are obtained so that the team can make strategic design changes before the final evaluation. For example, in the design for the last benchmark, the team had a 36:1 gear ratio of high torque, where when loaded, the car was unable to drive uphill in the given time frame. The results of these testing are indicative of design changes that the team must make to prevent underperformance at any event. However, the team should have run more tests on impact test, where the team did not foresee how easy it was for the coupler for the wheels to fracture.

Overall, the team enjoyed designing and testing the shifting mechanism, where there were practically no design restrictions. The team thrived under this creativity-encouraged environment, eventually proposed and realized a slide-shift mechanism that reached most goals. The most frustrating part of the project is having to pay attention to the budget, since any stock part from the storeroom was, under $20 budget of this project, always too expensive. For example, almost half of the budget ($28) went to the 6 bearings ($12) that the team thinks would facilitate sliding. In the future, the team suggests multiple standards under different budgets for students to choose from of their own will. For instance, while the events could remain the same, the difficulty could change depending on which tier of budget students decided to use. If the current events are to be inherited, suggestions will be made that the administrators of this class introduce more levels of similar events, where students under $40 get maximum scores if they did 1.5 m/s, 10 traversals, and 4kg load (subject to change), and under $60, they did more.

VI. Appendix

Appendix A. Author Contributions

Luke: Computed FEA Analysis and wrote FEA section. Added References to Appendix B.

Karthick: Machine Component Analysis, calculations for FEA, formatting, and overview

Theo: Reflection and calculations in appendix

Appendix B. Reference

[1] Autodesk, *Fusion 360,* Las Vegas: Autodesk, 2013.

[2] S. H. I. P. B. H. M. M. a. F. T. Mia Rismalia, "Infill pattern and density effects on the tensile properties of 3D printed PLA material," *Jounral of Pyhsics: Conference Series,* vol. 1402, no. 2, p. 044041, 2019.

[3]Adafruit, "DC Gearbox Moto- "TT Motor" - 200RPM- 3 to 6VDC," [Online]. Available: https://www.adafruit.com/product/3777. [Accessed December 2025].

Appendix C. Detailed Calculation

Gear Ratio calculation:

$$C_{wheel} = 2\pi R_{wheel} = 0.21\, m$$

Speed Event:

$$n_{1m/s} = 60 \cdot \frac{1}{C_{wheel}} = 284.2\, RPM$$

$$Gear\ Ratio_{speed} = \frac{n_{motor}}{n_{1m/s}} = 1.5 : 1$$

*Note here the motor RPM is the minimal rotation speed at stall torque, which means this would be the worst case scenario and the car in reality will run faster than this.

Strength Event:

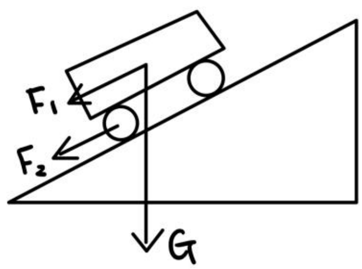


Figure: Free Body Diagram used for Force Analysis

Assuming:

$$T_{motor,\,stall} = 0.8\,kg \cdot cm = 0.078\,N \cdot m$$

$$n_{motor,\,stall} = 200$$

$$m_{full\,assembly} = 1.5\,kg$$

$$\mu_{rolling} = 0.02$$

$$F_1 = \sin\left(15^{\circ}\right) \cdot mg = 10.156\,N$$

$$F_2 = \mu_{rolling} mg = 0.758\,N$$

$$F_{wheel} = F_1 + F_2 = 10.914\,N$$

$$T_{wheel} = F_{wheel} \cdot R_{wheel} = 0.3667\,N \cdot m$$

$$Gear\ Ratio_{high\,torque} = \frac{T_{motor}}{T_{wheel}} = 4.6743{:}\,1$$

*Note here $Gear\ Ratio_{high\,torque} = \frac{T_{motor}}{T_{wheel}} = 4.6743$

Machine Component Analysis Calculations:

Motor Gear to Input Shaft

$$N1 = 12, N2 = 24, \frac{N1}{N2} = 2$$

Input Shaft to Intermediary "Shifter Shaft"

$$N3 = 12, N4 = 48, \frac{N4}{N3} = 4$$

2x4 = 8

Motor gear : gear of interest

$$8:1$$

Ft Calculation:

Input Torque: 0.8 kg.cm

Gear ratio = 8:1

$$T_{part\ of\ interest} = 8 * 0.8 = 6.4\ kg.cm$$

Convert to metric

$$1\ kg.cm = 0.09806\ N.m$$

T = 6.4*0.098066 = 0.6276 N.m

$$r = \frac{15\ mm}{2} = 0.0075\ m$$

$$F_t = \frac{T}{r} = \frac{0.6276}{0.0075} = 83.68\ N$$

If input torque =0.6 kg.cm

$$T_{part\ of\ interest} = 8 * 0.6 = 4.8\ kg.cm$$

T = 4.8*0.098066 = 0.470168 N.m

$$F_t = \frac{T}{r} = \frac{0.470168}{0.0075} = 62.76\ N$$

K_v calculation:

$$d = 0.015m * 3.28084 = 0.0492ft$$

$$v = \pi dn$$

$$v = \pi * (0.0492)(200)$$

Type equation here. $v = 30.94\ ft/min$

$$K_v = \sqrt{\frac{78 + \sqrt{30.94}}{78}} = 1.035$$

FEA Load Calculation:

Assume worst case at stall torque, 0.8 kg.cm.

Load on gear on input first shaft with smallest gear has gear ratio 1:2, so 1.6 kg.cm.

$$r = 0.0075\ m$$

$$F = \frac{1.6 * 0.0980665}{0.0075} = 20.9\ N$$